MIL-M-46366A(MU)
28 December 1964
SUPERSEDING
MIL-M-46366(MU)
8 February 1963

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M90

1. SCOPE

1.1 This specification covers a fire control instrument which provides cant correction and boresight adjustment for the direct fire telescope which it supports (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F8299670	Mount, Telescope: M90
IEL 8299670	List of Inspection Documents for Mount, Telescope: M90

FSC 1240

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PACKAGING DATA SHEET

MIL-P-14232/P8299670 Packaging of Mount, Telescope: M90

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing F8299670 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental service conditions.

3.3.1 Standard ambient temperature.- The mount shall meet the requirements of 3.3.3 to 3.8 inclusive, and the applicable requirements of 3.9, at a standard ambient temperature between +60° and +90° Fahrenheit (F).

3.3.2 Extreme operating temperatures.- The mount shall meet the applicable requirements of 3.9 while exposed to, and thermally stabilized at, ambient temperatures of -40° and +150°F.

3.3.3 Vibration.- The mount shall withstand a vibration of 30 ± 5 cycles per second frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds (see 3.4). Subsequent to vibration the mount shall show no evidence of damage or physical failure, and it shall meet the requirements of 3.4.2 to 3.9 inclusive.

3.4 Alignment.

3.4.1 Pre-vibration.- When the mount is positioned as shown on Drawing F8299670, the following conditions shall be met:

(a) The geometric axis of the socket section which receives the telescope shall be horizontal.

(b) The geometric axis of the socket section shall be parallel to the flat mounting surface and the base of the mounting keyway.

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3.4.2 Post-vibration.- Subsequent to vibration, the alignment of the socket section shall not have shifted from the position specified in 3.4.1 in excess of 0.4 mil.

3.5 Azimuth boresight adjustment.- Rotation of the azimuth boresight adjustment worm shall deflect the geometric axis of the mount's socket section in a horizontal plane a minimum of 11.0 mils to the left and 11.0 mils to the right of the alignment position of 3.4.1.

3.6 Elevation boresight adjustment.- Rotation of the elevation boresight adjustment worm shall elevate and depress the geometric axis of the socket section in a vertical plane a minimum of 11.0 mils up and 11.0 mils down from the alignment position of 3.4.1.

3.7 Cant correction.- Rotation of the cant correction knob shall rotate the socket section about its geometric axis a minimum of 20 degrees to the right and 20 degrees to the left of the position established in 3.4.1. Alignment of the geometric axis shall not change in excess of 0.7 mil from the position established in 3.4.1.

3.8 Backlash.- Backlash in the cant correction mechanism shall not exceed 1/4 inch movement as measured on the circumference of the cant correction knob.

3.9 Operability.- The torque, in pound-inches or ounce-inches as indicated, to rotate the controls at the standard ambient temperature of 3.3.1 and the extreme operating temperatures of 3.3.2 shall be in accordance with the maximum and minimum requirements of Table I.

TABLE I
OPERABILITY

Control	-40°F	+150°F	Standard Ambient Temperature
	Torque (lb-in.)	Torque (lb-in.)	Torque
Azimuth adjustment	10 max.	5 max.	8 oz-in. min.
Elevation adjustment	10 max.	5 max.	8 oz-in. min.
Cant correction knob	12 max.	12 max.	8 lb-in. max.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 8299670. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105 and the inspection level and sampling plans specified in Table II, titled "Classification of Defects". Examinations and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.5%		
101. Physical failure, post vibration	3.3.3	4.5.2

4.

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	REQUIREMENT	TEST PROCEDURES
102. Azimuth boresight	3.5	4.6.3
103. Elevation boresight	3.6	4.6.4
104. Cant correction	3.7	4.7
105. Backlash	3.8	4.8
106. Operability	3.9	4.9

MINOR: AQL 4.0%

201. Paint and burrs on locating and bearing surfaces	3.2	Visual

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control test.

4.3.1 General.- One mount shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Visual - As per drawings and specifications
302. General specification	3.2	Visual; MIL-F-13926

4.3.2 Environmental.- Three mounts shall be selected at random as control samples from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirement and test:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303. Operability at extreme temperatures	3.3.2	4.5.1

4.3.3 Functional.- Three mounts shall be selected at random as control samples from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirement and test:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
304. Standard ambient temperature	3.3.1	4.9

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4.3.4 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents, IEL 8299670.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test methods and procedures" specified in 4.5 and with the "Component inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 Extreme operating temperatures.- This test shall be performed by exposing the mount to the temperatures indicated in 3.3.2 by means of a cold chest and oven that are capable of maintaining these temperatures until the mount is thermally stabilized. While at these temperatures, the azimuth and elevation adjusting worms and the cant correction knob shall be tested for torque in each direction, using commercial torque-testing equipment. The torque readings shall not exceed the applicable requirements of Table I when the controls are operated in both the clockwise and counterclockwise directions.

6.

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4.5.2 Vibration.- The mount shall be vibrated on a vibrating machine, Drawing F7560095, with adapter F7680691, for the amplitude specified in 3.3.3 within 0.010 inch, at the number of cycles and time duration specified. Subsequent to vibration the mount shall then be subjected to a visual-tactile examination for damage and loose parts and shall be tested in accordance with 4.6.2 to 4.9 inclusive.

4.6 Alignment.

4.6.1 Pre-vibration.- The mount shall be aligned on special test equipment, Drawing F7681052, which shall position the mount as specified in 3.4.1.

4.6.2 Post-vibration.- The mount shall be positioned in the special test fixture noted in 4.6.1. The alignment of the socket section shall not have shifted more than the tolerance specified in 3.4.1.

4.6.3 Azimuth boresight adjustment.- The mount shall be positioned as specified in 3.4.1. With the aid of an auxiliary telescope, positioned in the socket of the mount, and a target graduated in mils, the minimum horizontal movement from center position, when turning the azimuth adjusting screw to the left and to the right, shall meet the requirements of 3.5.

4.6.4 Elevation boresight adjustment.- This test is performed by using the same method utilized for the azimuth adjustment test in 4.6.3, except this test is made by turning the elevation adjusting screw for vertical movement of the mount socket. Elevation and depression movement shall meet the requirements of 3.6.

4.7 Cant correction.- With the mount aligned as specified in 3.4.1, sight through the auxiliary telescope that is positioned in the mount socket and observe a target that indicates left and right 20-degree angles and a 0.7 mil radius circle. Rotate the cant correction knob so the mount socket rotates the telescope around its geometric axis. There shall be no movement in excess of the requirements of 3.7.

4.8 Backlash.- With the mount positioned as specified in 3.4.1, rotate the cant correction knob in one direction to center the geometric axis of the mount socket to the center of the target. When centered, mark coinciding lines on the knob and the boss. Continue rotation of the cant knob in the same direction for at least a quarter of a turn, then reverse direction back to the center position on the target without any override. Backlash shall be measured as the difference between the marked lines on the knob and the boss, and shall not exceed the requirements of 3.8.

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4.9 Operability.- The azimuth and elevation adjustment worms and the cant correction knob shall meet the torque requirements of Table I, at the temperatures specified in 3.3.1 and 3.3.2, when tested with a torque wrench and an adapter shown on Drawings C8572319 and C7680692, respectively.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/PB299670. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Telescope Mount: M90 is used to support Elbow Telescopes: M92D and M92F for direct fire of the 106mm recoilless rifle.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

Custodian:

Army - MU

Preparing activity:

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Project No. 1240-A440Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner, and send to preparing activity.*

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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